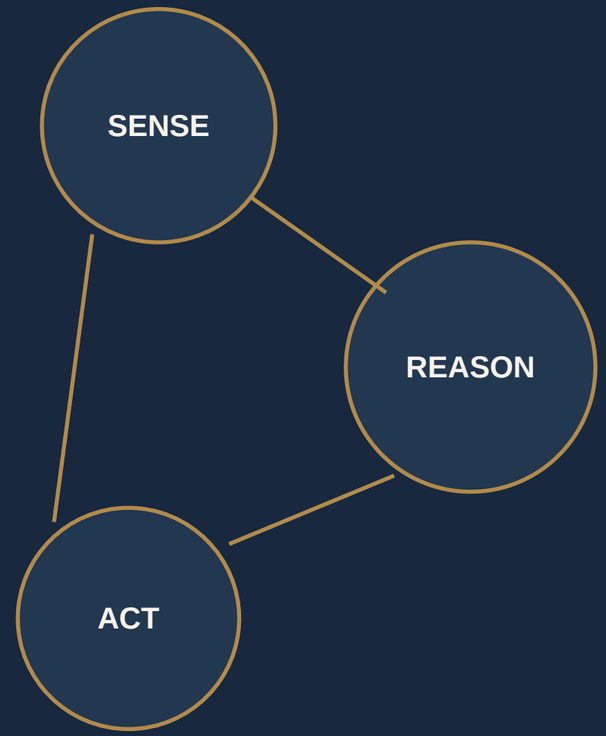
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Introduction to Artificial Intelligence

AI.101 | Foundations, ideas and intelligent systems

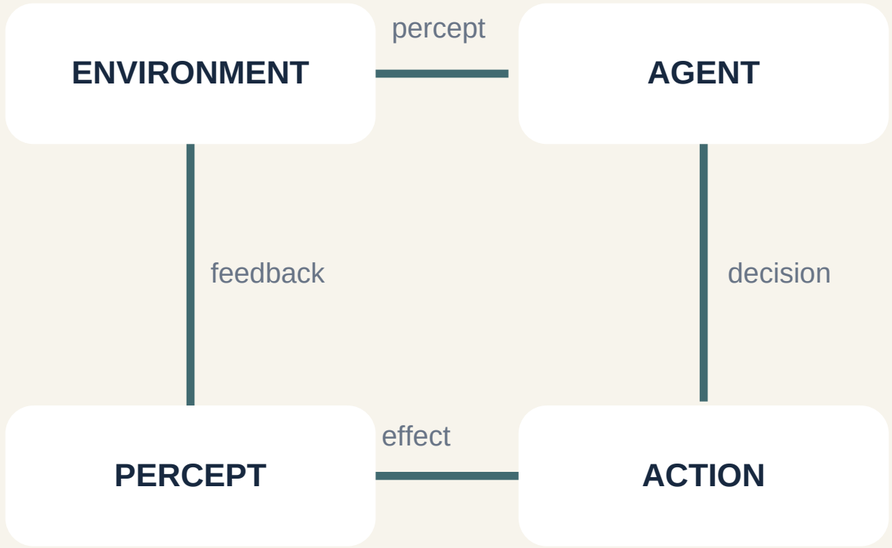


What makes a system intelligent?

Begin with behavior, not the label.

A useful starting point

An intelligent system perceives its environment, selects actions and adapts its behavior to achieve goals.



By the end of this course, you can…

- Explain major traditions in AI and how they complement one another.
- Formulate an agent, search or reasoning problem with explicit assumptions.
- Compare learning approaches using evidence, limitations and engineering trade-offs.
- Build and evaluate a small intelligent system, then communicate its behavior responsibly.

AI is a field with several intellectual lineages

Ideas evolve; they do not simply replace one another.

Through-line: represent a problem → reason or learn → act → evaluate the result.

Four traditions. Different strengths.

Symbolic

Explicit rules, logic and structured knowledge. Strong when explanations and constraints matter.

Statistical

Models infer patterns from data. Strong when uncertainty and variation are central.

Connectionist

Neural networks learn distributed representations. Powerful for perception and language.

Hybrid

Combine learned components with rules, search or structured models.

Agent design begins with the task environment

- Performance measure: what counts as success?
- Environment: what can change, and what is hidden?
- Actuators: what actions are available?
- Sensors: what observations can the system receive?

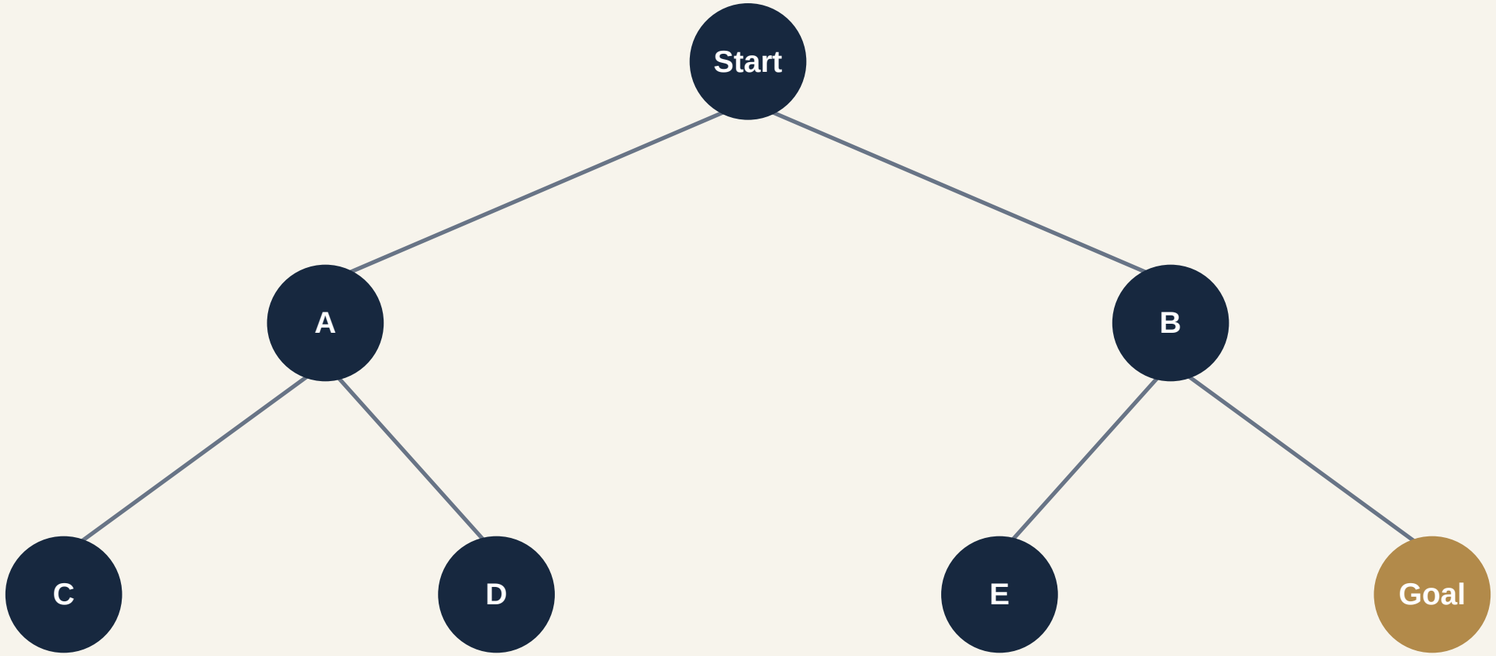
Example · delivery robot

Success balances safe arrival, time, energy and accessibility—not just shortest distance.

GOAL SENSORS ACTIONS FEEDBACK

Search turns choices into a structured space

A search algorithm explores possible states toward a goal.



Breadth-first search: expand by depth. A* search: combine path cost with an estimate to goal.

Representations determine what can be reasoned about

Facts

Store observations or asserted truths with provenance and confidence.

Rules

Encode relationships and conditions; trace how conclusions follow.

Uncertainty

Represent incomplete evidence; update beliefs rather than pretending certainty.

Representation → inference → decision

Learning is an evidence pipeline—not a magic box

DATA
examples

FEATURES
useful signal

MODEL
fit parameters

TEST
unseen cases

DEPLOY
monitor drift

Ask: what data was available, what was held out, and what errors matter in context?

One field, many kinds of systems

Perception

Vision, speech and sensing turn signals into useful representations.

Language

Search, translation and dialogue systems process and generate text.

Action

Robots and autonomous systems choose actions under constraints.

Generative tools

Models produce new content; quality and provenance need evaluation.

Decision support

Predictions inform people; responsibility and oversight remain essential.

Lab · build a small goal-directed agent

From specification to evidence

- Define a grid world, start, goal and obstacles.
- Implement a baseline random agent, then breadth-first search.
- Measure path length, nodes expanded and success rate.
- Change obstacles; explain when the first approach fails.

Check your understanding

Discuss

Can a system be useful without being intelligent? What evidence would change your answer?

Apply

Choose one campus or community problem. Specify its goal, observations, actions and a measure of success.

Next: AI.101 continues with problem formulation and search strategies.

Remember: define the task, make assumptions visible, and evaluate behavior—not hype.